

UNIT 4

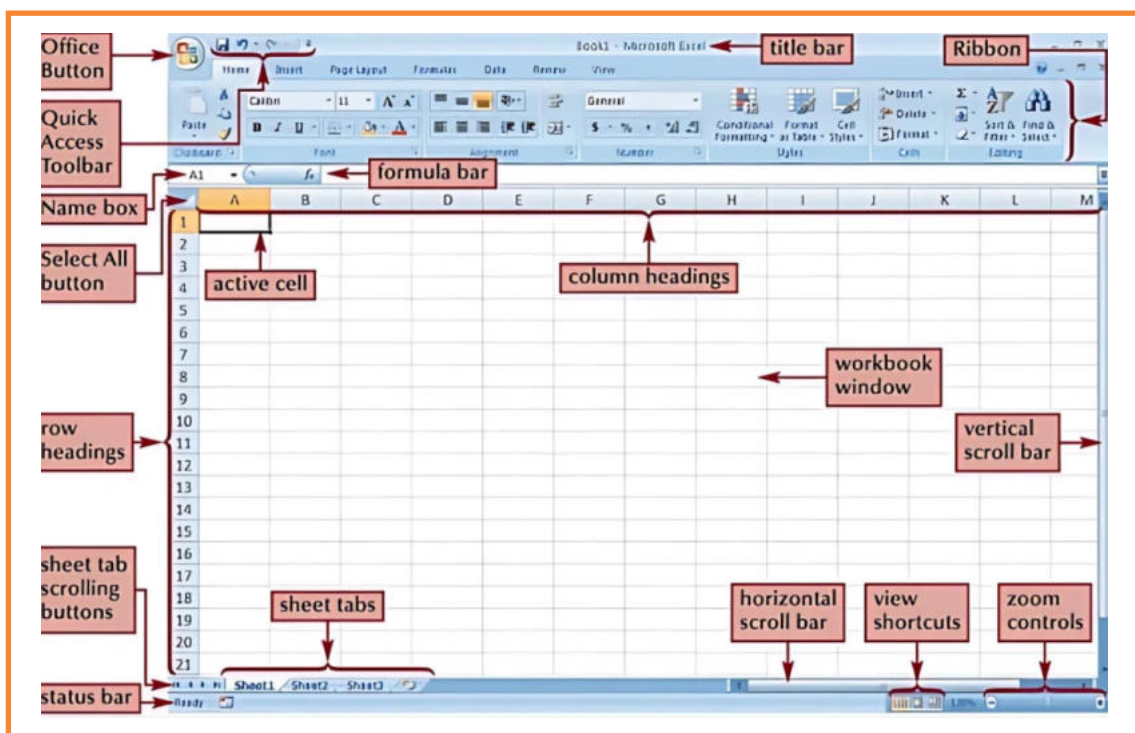
EXPLORING MS EXCEL FOR DATA MANAGEMENT



Learning Objectives

1. Create charts like column and bar charts.
2. Understand how to use cell references, including relative, absolute, and mixed references.
3. Explore built-in functions and learn how to insert and apply them in MS Excel.
4. Learn how to sort data in MS Excel

Microsoft Excel is a powerful spreadsheet application that is perfect for maintaining long lists of data, budgets, sales figures, and other records. In this chapter, we will explore important features of Excel such as creating tables, charts, using cell references, and applying formulas. Let's discuss these features step-by-step.

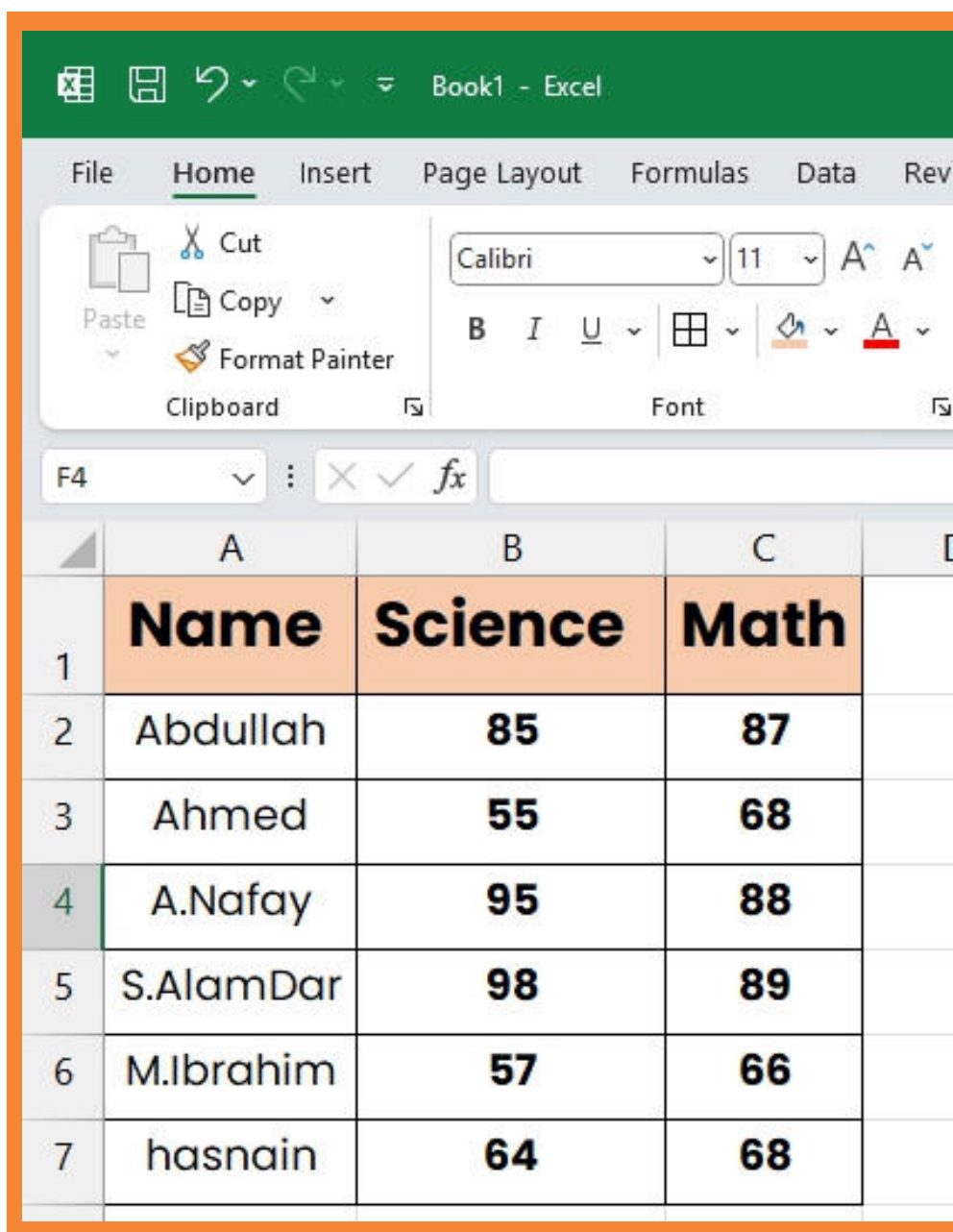


Creating a Bar or Column Chart

To create a Column or Bar Chart in Excel, let's use a simple example of marks secured by students in Science and Math.

Preparing Chart Data

The data below shows marks scored by students:

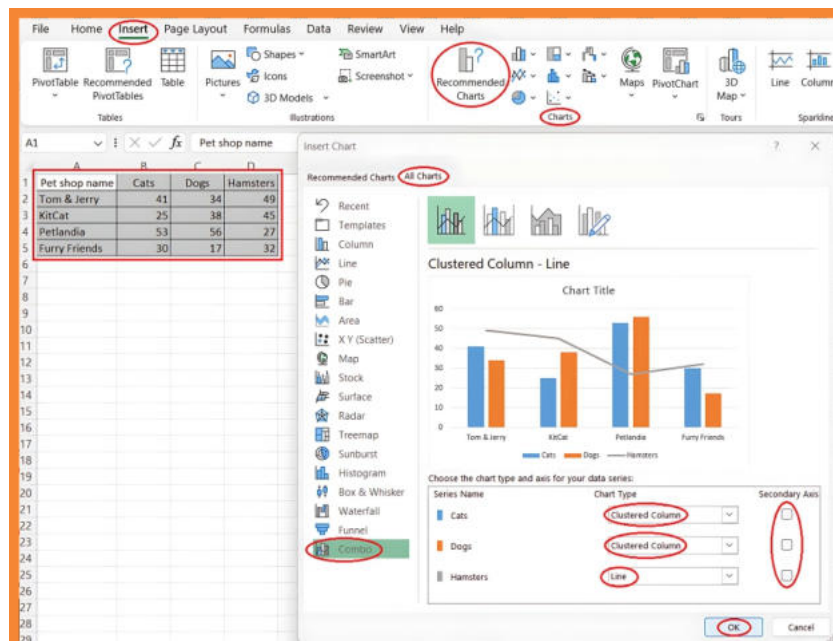


The screenshot shows the Microsoft Excel interface with the 'Home' tab selected. The ribbon includes options for Clipboard (Paste, Cut, Copy, Format Painter), Font (Calibri, size 11, bold, italic, underline, text color, background color), and Cells (Merge & Center, Wrap Text, Orientation, Fill Color, Border). The active cell is F4. The data table is as follows:

	A	B	C
1	Name	Science	Math
2	Abdullah	85	87
3	Ahmed	55	68
4	A.Nafay	95	88
5	S.AlamDar	98	89
6	M.Ibrahim	57	66
7	hasnain	64	68

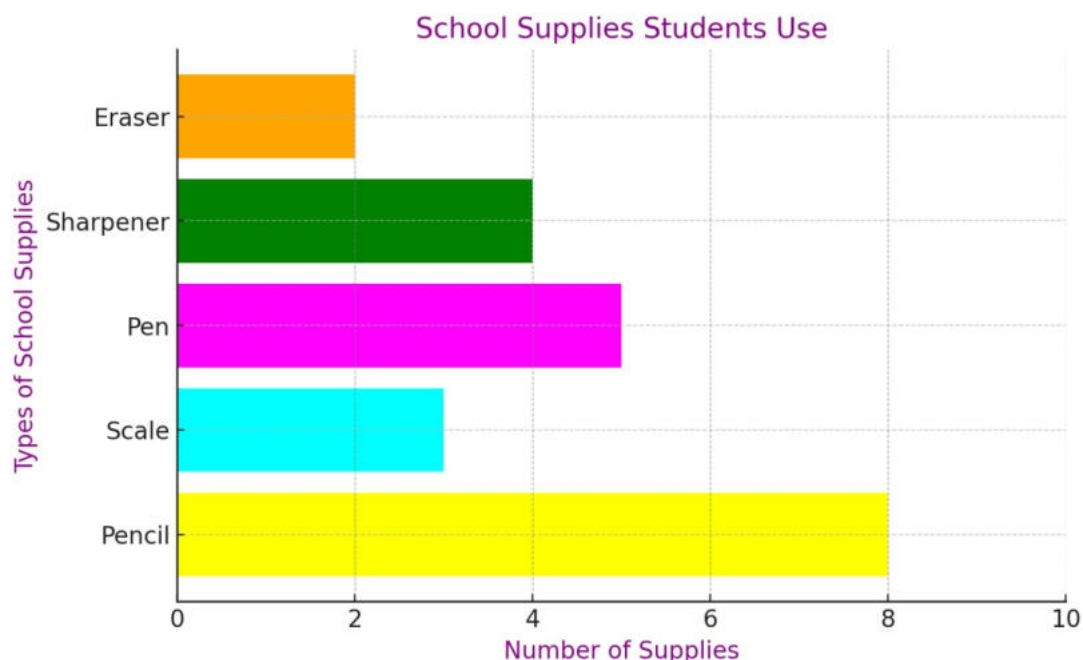
Creating Your Column Chart

1. Select your data range, e.g., B5:D13.
2. Go to the **Insert** menu and click on the **Column Chart** button.
3. Choose the 3D Clustered Column Chart option.
4. Excel will automatically generate the chart for your data.



Creating Your Bar Chart

To create a Bar Chart, follow the same steps as for the Column Chart, but select the "Bar Chart" option instead of "Column Chart."



Absolute and Relative References in Excel Formulas

Excel uses three types of cell references in formulas, each with specific purposes and functionalities. Let's break it down step-by-step:

Step 1: Understanding Relative References

- A **relative reference** changes dynamically when a formula is copied to another location.
- Example: If a formula in cell C2 is `=A2 + B2`, copying it to cell C3 will automatically adjust the formula to `=A3 + B3`.

Try it Yourself:

1. In cell C2, enter `=A2 + B2`.
2. Copy the formula to C3 and observe the adjustment to `=A3 + B3`.

Step 2: Understanding Absolute References

- An **absolute reference** is fixed and does not change when the formula is copied. It uses a dollar sign (\$) before the column and row (e.g., `$A$2`).
- Example: To multiply all cells in column A by the value in cell B1, use the formula `=A2*$B$1` in cell C2.

Try it Yourself:

1. Enter a value in cell B1.
2. In cell C2, enter `=A2*$B$1` and copy the formula to other rows. Notice how `$B$1` remains constant.

Step 3: Understanding Mixed References

- A **mixed reference** combines relative and absolute references. Either the column is fixed (e.g., `$A2`) or the row is fixed (e.g., `A$2`).
- Example: In the formula `=A$2 * $B3`, the row for `A$2` remains constant, while the column for `$B3` stays fixed when copied.

Try it Yourself:

1. In cell D2, enter `=A$2 * $B3`.
2. Copy the formula across rows and columns to observe how each part remains fixed.

	A	B	C
1	Item	Price, USD	Price, EUR
2	Apples	\$5.00	€4.65
3	Avocados	\$4.50	
4	Bananas	\$3.90	
5	Grapes	\$9.90	
6	Lemons	\$4.70	
7	Pears	\$2.40	
8	Watermelon	\$2.50	

Built-in Functions in MS Excel

Excel provides a variety of built-in functions to simplify and automate calculations. These functions are categorized for easy access:

Text Functions

- **LOWER:** Converts text to lowercase.
- **UPPER:** Converts text to uppercase.
- **CONCATENATE:** Joins text from multiple cells into one string.
- **LEFT:** Extracts a specified number of characters from the start of a string.
- **RIGHT:** Extracts a specified number of characters from the end of a string.

Date & Time Functions

- **NOW:** Returns the current date and time.
- **TODAY:** Returns the current date only.
- **TIME:** Returns a time value based on specified hours, minutes, and seconds.

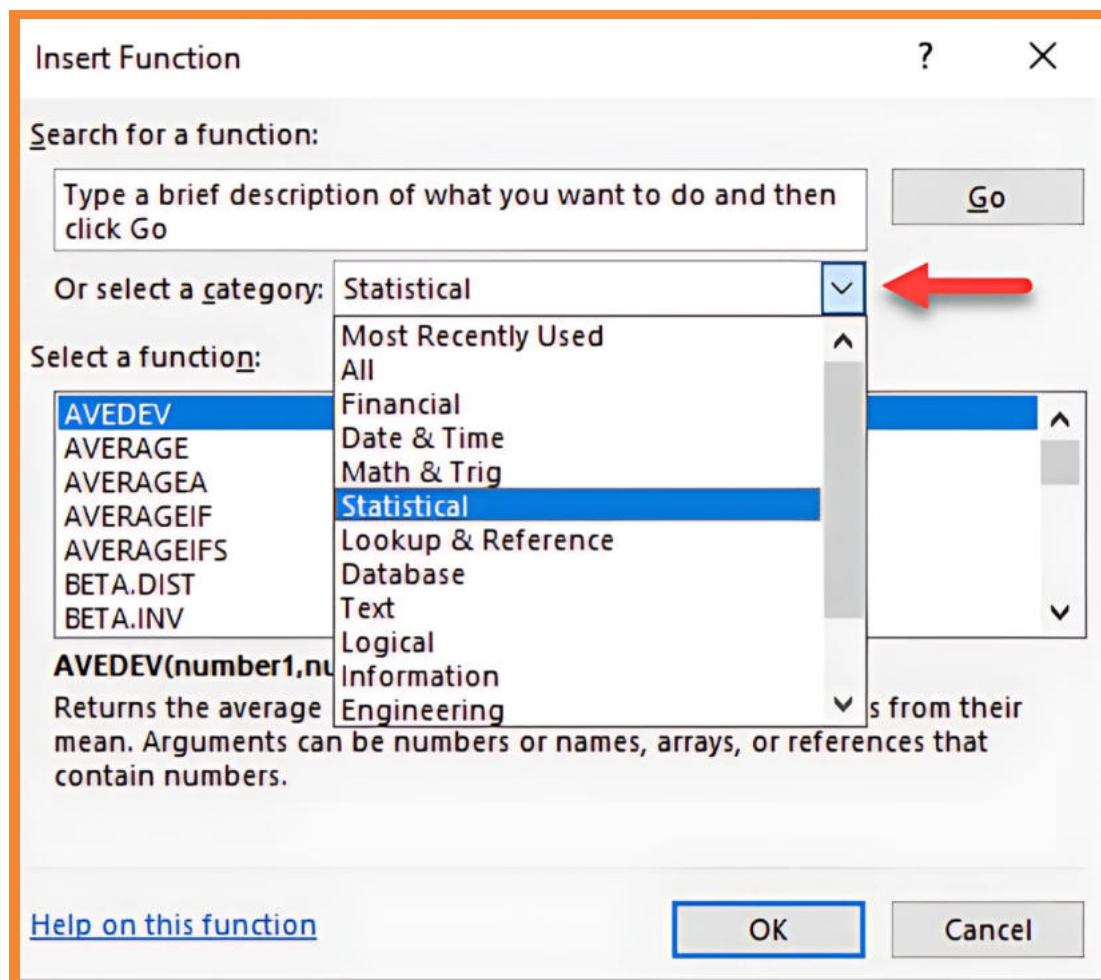
Statistical Functions

- **SUM:** Adds up values in a range..

- **MAX:** Finds the largest value in a range.
- **MIN:** Finds the smallest value in a range.
- **COUNTIF:** Counts the number of cells that meet a specific condition.

Logical Functions

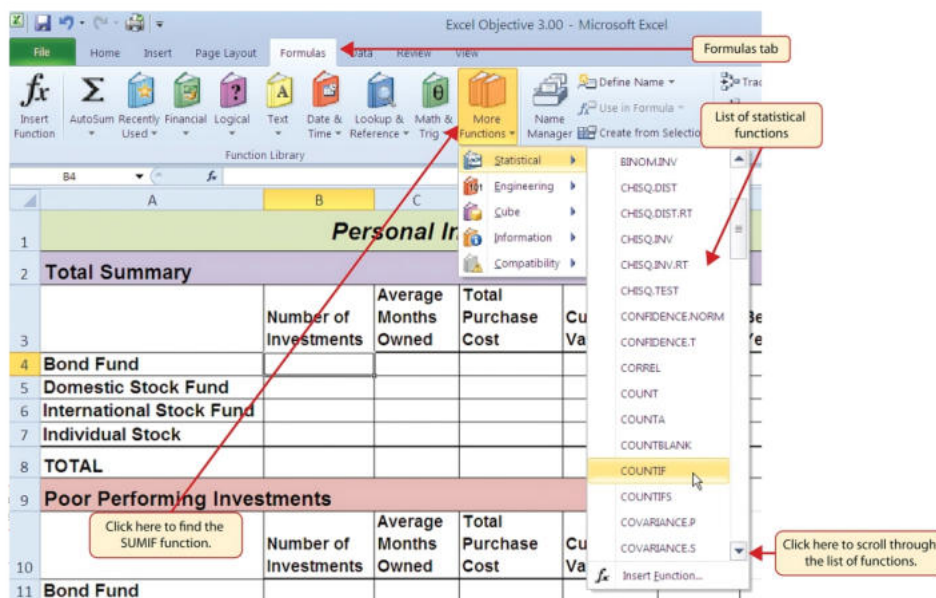
- **IF:** Returns a value based on a condition (e.g., =IF(A1>10, "Yes", "No")).
- **AND:** Checks if multiple conditions are true.
- **OR:** Checks if at least one condition is true.
- **NOT:** Reverses the logical value of a condition



How to Insert a Function in Excel

Functions are powerful tools in Excel to perform automated calculations. Here's how to use them step-by-step:

- 1. Select a Cell:** Click the cell where you want the result of the function.
- 2. Insert Function Button:** Click on the "Insert Function" button (fx) located next to the formula bar.
- 3. Choose a Function:** In the dialog box, search for a function or select one from a category. For example, select COUNTIF from the Statistical category.
- 4. Define Arguments:** Enter the required inputs (range, criteria, etc.) in the "Function Arguments" box.
- 5. Click OK:** The result of the function will appear in the selected cell.



Example: Use the COUNTIF function to count cells with values greater than 5. Select the range A1:C2, and enter the condition "> 5" in the criteria box. Excel will count and display the result.

Sorting Data in Excel

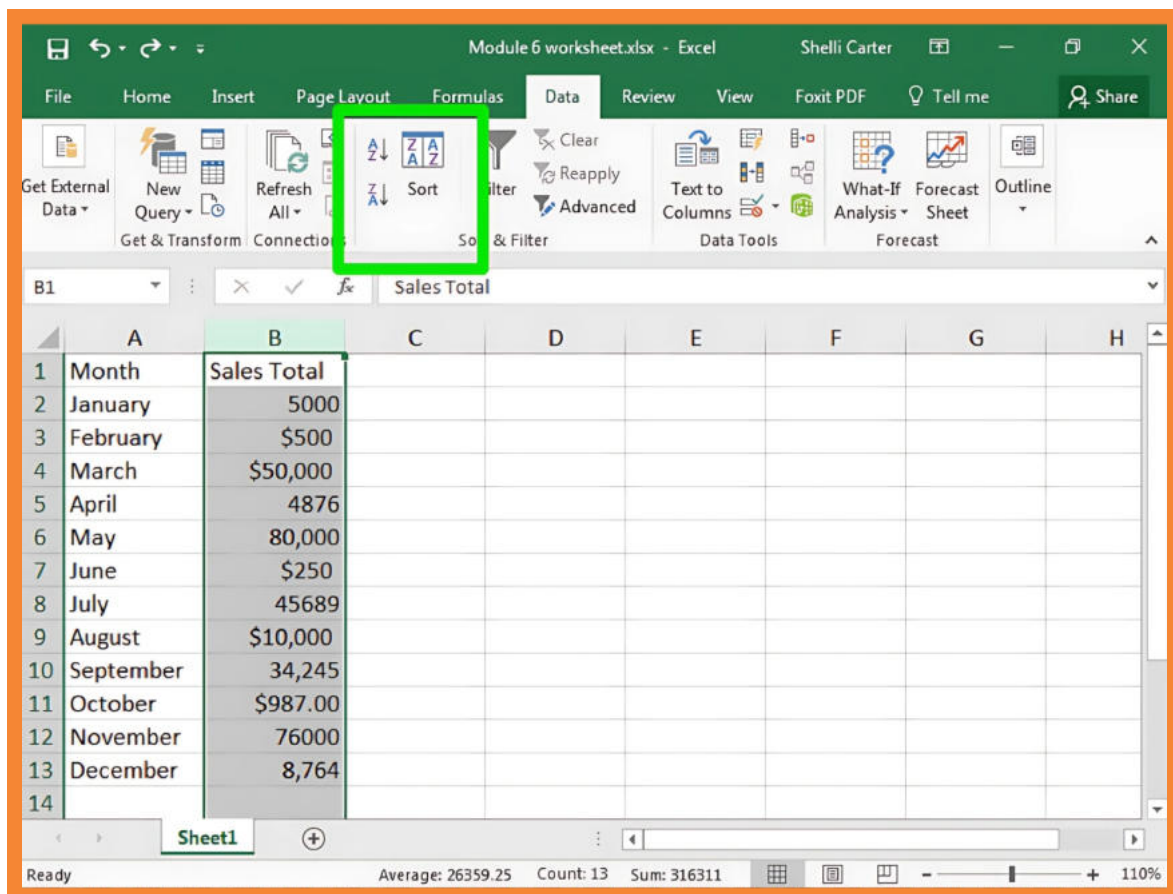
Sorting data helps organize it effectively.

• Sorting Data Alphabetically:

1. Select a cell in the column you want to sort (e.g., Name).
2. Go to the **Data** tab and click "Sort A to Z."

• Sorting Data Numerically:

1. Select a cell in the column (e.g., Science).
2. Click "Sort Smallest to Largest."



MS Excel is a versatile tool for organizing, analyzing, and visualizing data. By mastering charts, functions, sorting, and formulas, you can efficiently manage data for school, work, or personal projects.

EXERCISES

01: Choose the correct answer.

1. Which function is used to convert text to uppercase in Excel?

- a) LOWER
 - b) CONCATENATE
 - c) UPPER
 - d) RIGHT
-

2. What type of reference is \$A\$1?

- a) Relative
 - b) Absolute
 - c) Mixed
 - d) None of the above
-

3. What is the shortcut to insert a function in Excel?

- a) Alt + F4
 - b) Ctrl + F
 - c) fx Button
 - d) Ctrl + Shift
-

4. What type of chart presents data in vertical columns?

- a) Bar Chart
- b) Column Chart
- c) Pie Chart
- d) Line Chart

5. Which function returns the current date and time?

- a) TODAY
- b) NOW
- c) TIME
- d) DATE

6. In Excel, what does the \$ sign in a reference indicate?

- a) Shortcut
- b) Locked reference
- c) Deleted column
- d) None of the above

7. To copy a formula without changing references, which reference type is used?

- a) Relative
- b) Mixed
- c) Absolute
- d) Conditional

8. Which tab is used to create charts in Excel?

- a) Home
 - b) Data
 - c) Insert
 - d) Review
-

9.What is the default function to add up a range of values?

- a) MAX
- b) COUNTIF
- c) SUM
- d) LEFT

10.What is the term for a chart presenting data in horizontal bars?

- a) Column Chart
- b) Bar Chart
- c) Pie Chart
- d) Scatter Chart

02: Fill in the blanks:

1. A _____ reference adjusts dynamically when copied to another cell.
2. To lock a cell reference, use the _____ symbol.
3. The _____ function returns the largest value in a range.
4. _____ charts display data in vertical columns.
5. The _____ function joins text from multiple cells.
6. The _____ button is used to insert a function in Excel.

03: Select True or False for each statement.

	TRUE	FALSE
Relative references remain constant when copied.	☐	☐
The TODAY function returns the current time.	☐	☐
You can use mixed references to fix either the row or column in a formula.	☐	☐
Bar charts display data in vertical columns.	☐	☐
COUNTIF is a logical function.	☐	☐
Excel functions can automate calculations.	☐	☐

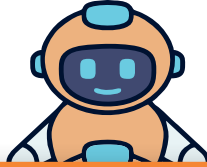
04: Answer the following questions briefly.

1. Explain the difference between absolute and relative references.
2. What is the purpose of the UPPER function?
3. How do you sort data alphabetically in Excel?
4. What is the function of the \$ symbol in Excel formulas?
5. What steps do you follow to insert a function in Excel?
6. Define the term "mixed reference".

05: Provide detailed answer for the following questions.

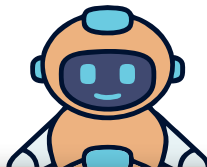
1. Explain the process of creating a Column Chart in Excel with an example.
2. Describe three types of cell references with examples.
3. What are built-in functions in Excel? Describe three categories of functions.
4. How do you use the COUNTIF function? Provide an example.

Lab Activity 1: Creating Charts



- Create a data table for five students showing their marks in three subjects.
- Generate both a Column Chart and a Bar Chart to visualize the data.
- Save and label the charts appropriately.

Lab Activity 2: : Using Functions and References



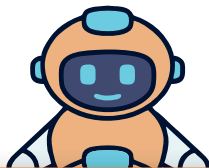
- Enter a list of 10 numbers in a column.
- Use SUM, MAX, and MIN functions to calculate the total, maximum, and minimum values.
- Apply relative, absolute, and mixed references in different formulas to calculate totals using specific criteria.

Lab Activity 3: Working with Built-in Functions



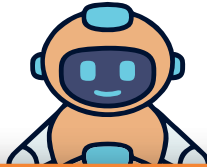
- Enter a list of employee names and their monthly salaries in two columns.
- Use the following functions:
- **SUM** to calculate the total salary.
- **AVERAGE** to calculate the average salary.
- **MAX** to find the highest salary.
- **MIN** to find the lowest salary.
- Save the file and name it "SalaryCalculations.xlsx."

Lab Activity 4: Sorting and Filtering Data



- Create a data table of 15 products with their prices and stock availability (In Stock/Out of Stock).
- Sort the products alphabetically.

Lab Activity 5: Using Absolute and Mixed References



- Create a data table showing the price of 5 products and the quantity purchased.
- Use a formula with an absolute reference to calculate the total cost (price * quantity).
- Add a tax percentage in a separate cell and use a mixed reference to calculate the total cost after tax.
- Save the file as "TotalCostCalculations.xlsx."